

Model 1: Abundance Plots

This file contains the processing code for Model 1: Comparing Temporally Blind and Temporally Explicit Metrics. Before running this file, you must have extinction time and population size outputs from running Model 1 for both TPCs using Model1.nb; edit import and export locations (via 'saveto') for your own device. Use to generate Fig 4.

```
saveto = "/SAVETO/";
```

Load Data

Run import code before creating any figures. Edit import/export file names/locations for your local system and, if necessary, edit loaded parameter space to match your imported run. Default size is tmax2000_dims21,4,1,1000, corresponding with 1000 runs of 2000 time steps at each of 21 autocorrelation levels, 4 mean temperatures, and 1 standard deviation of temperature.

```
In[2]:= extTime = Import[saveto <> "extMetrics8_extTime_tmax2000_dims21,4,1,1000.m", "MX"];
p = Import[saveto <> "extMetrics8_popsizes_tmax2000_dims21,4,1,1000.m", "MX"];

extTimeT =
  Import[saveto <> "extMetrics8_trop_extTime_tmax2000_dims21,4,1,1000.m", "MX"];
pT = Import[saveto <> "extMetrics8_trop_popsizes_tmax2000_dims21,4,1,1000.m", "MX"];

tmax = 2000; (*2000*)
maxsims = 1000; (*1000*)
γvals = Range[0, 2, .1]; (*21*)
μTvals = Range[23, 26, 1]; (*4*)
σTvals = Range[3, 3, 1]; (*1*)
```

Calculate Temporally Blind Metrics

Calculate the expected population abundances given the means and standard deviation of temperature without any input from the simulations.

```
In[11]:= (*TEMPORATE - TPC1*)
gain[T_] := Exp[-((T - TI)^2)/β];
loss[T_] := ma Exp[mb T] + mc;
r[T_] := (gain[T] - loss[T]) / .755;

(* Parameters of the TPC model *)
```

```

TI = 29;
 $\beta$  = 180;
ma = 0.000005;
mb = 0.4;
mc = 0.05;

 $\alpha$  = .001;

(*array of temperatures for each parameter set*)
W = ConstantArray[0, {Length[ $\mu$ Tvals], Length[ $\sigma$ Tvals], tmax}];
Do[W[[i, j, k]] = InverseCDF[NormalDistribution[ $\mu$ Tvals[[i]],  $\sigma$ Tvals[[j]]], k * (1 / tmax)],
  {i, 1, Length[ $\mu$ Tvals]}, {j, 1, Length[ $\sigma$ Tvals]}, {k, 1, tmax - 1}]
Do[W[[i, j, tmax]] =
  InverseCDF[NormalDistribution[ $\mu$ Tvals[[i]],  $\sigma$ Tvals[[j]]], (tmax - .5) * (1 / tmax)],
  {i, 1, Length[ $\mu$ Tvals]}, {j, 1, Length[ $\sigma$ Tvals]}];

(*set initial pop size to mean pop size for each parameter set*)
ninit = ConstantArray[0, {Length[ $\mu$ Tvals], Length[ $\sigma$ Tvals]}];
Do[ninit[[i, k]] = (Mean[Table[r[W[[i, k, j]]], {j, 1, tmax}]]]) /  $\alpha$ ,
  {k, 1, Length[ $\sigma$ Tvals]}, {i, 1, Length[ $\mu$ Tvals]}]
N[ninit]

l23 = ConstantArray[ninit[[1]], Length[ $\gamma$ vals]];
l23t = Transpose[{ $\gamma$ vals, Flatten[l23]}];
l24 = ConstantArray[ninit[[2]], Length[ $\gamma$ vals]];
l24t = Transpose[{ $\gamma$ vals, Flatten[l24]}];
l25 = ConstantArray[ninit[[3]], Length[ $\gamma$ vals]];
l25t = Transpose[{ $\gamma$ vals, Flatten[l25]}];
l26 = ConstantArray[ninit[[4]], Length[ $\gamma$ vals]];
l26t = Transpose[{ $\gamma$ vals, Flatten[l26]}];

(*TROPICAL - TPC2*)
gain[T_] := Exp[-((T - TI) ^ 2) /  $\beta$ ];
loss[T_] := ma Exp[mb T] + mc;
r[T_] := (gain[T - 2] - loss[T - 2]) / .4067;

(* Parameters of the TPC model *)
TI = 35;
 $\beta$  = 130;
ma = 0.000005;
mb = 0.4;
mc = 0.05;

 $\alpha$  = .001;

```

```

(*array of temperatures for each parameter set*)
W = ConstantArray[0, {Length[μTvals], Length[σTvals], tmax}];
Do[W[[i, j, k]] = InverseCDF[NormalDistribution[μTvals[[i]], σTvals[[j]]], k * (1 / tmax)],
  {i, 1, Length[μTvals]}, {j, 1, Length[σTvals]}, {k, 1, tmax - 1}]
Do[W[[i, j, tmax]] =
  InverseCDF[NormalDistribution[μTvals[[i]], σTvals[[j]]], (tmax - .5) * (1 / tmax)],
  {i, 1, Length[μTvals]}, {j, 1, Length[σTvals]}];

(*set initial pop size to mean pop size for each parameter set*)
ninit = ConstantArray[0, {Length[μTvals], Length[σTvals]}];
Do[ninit[[i, k]] = (Mean[Table[r[W[[i, k, j]]], {j, 1, tmax}]] / α,
  {k, 1, Length[σTvals]}, {i, 1, Length[μTvals]}]

```

```
N[ninit]
```

```

lt23 = ConstantArray[ninit[[1]], Length[γvals]];
lt23t = Transpose[{γvals, Flatten[lt23]}];
lt24 = ConstantArray[ninit[[2]], Length[γvals]];
lt24t = Transpose[{γvals, Flatten[lt24]}];
lt25 = ConstantArray[ninit[[3]], Length[γvals]];
lt25t = Transpose[{γvals, Flatten[lt25]}];
lt26 = ConstantArray[ninit[[4]], Length[γvals]];
lt26t = Transpose[{γvals, Flatten[lt26]}];

```

```
Out[25]=
{{851.715}, {845.332}, {797.846}, {691.715}}
```

```
Out[48]=
{{377.737}, {444.963}, {497.597}, {520.587}}
```

Process Simulation Data

Given the imported extinction times and population sizes, calculate the extinction metrics (abundance, stability, and probability of extinction) and create plotting arrays.

```

In[57]:= (*initialize functions for calculating the proportion variance index (PV)
and consecutive disparity index (CD); only the latter is computed here*)
CDindex[data_List] := Module[{n, diffs}, n = Length[data];
  diffs = Abs[Log[data[[2 ;;]] / data[[1 ;; -2]]]];
  Mean[diffs]]
(*PVindex[data_List] := Module[{n, pairs, zsum}, n = Length[data];
  pairs = Subsets[data, {2}];
  zsum = Total[(1 - (Min[#] / Max[#])) & /@ pairs];
  (2 * zsum) / (n * (n - 1))] *)

```

```

(*TEMPERATE*)
meansMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stdevsMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stabilityMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
extMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
cdMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];

(*averaging over all timesteps before extinction within a simulation*)
Do[If[extTime[[i, j, 1, l]] < tmax, extMatrix[[i, j, l]] = 1, extMatrix[[i, j, l]] = 0];
  meansMatrix[[i, j, l]] = Mean[p[[i, j, 1, l, 2 ;; extTime[[i, j, 1, l]]]]];
  stdevsMatrix[[i, j, l]] =
    StandardDeviation[p[[i, j, 1, l, 2 ;; extTime[[i, j, 1, l]]]]];
  stabilityMatrix[[i, j, l]] = meansMatrix[[i, j, l]] / stdevsMatrix[[i, j, l]],
  {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

Do[cdMatrix[[i, j, l]] = CDindex[p[[i, j, 1, l, 2 ;; extTime[[i, j, 1, l]]]] + .1],
  {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]
Do[cdMatrix[[i, j]] = DeleteAnomalies[cdMatrix[[i, j]],
  {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*averaging across simulations with the same params*)
meansVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ1 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ3 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stdevsVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stabilityVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
propExt = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ1 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ3 = ConstantArray[0, {Length[γvals], Length[μTvals]}];

Do[meansVals[[i, j]] = Mean[meansMatrix[[i, j]]];
  meansQ1[[i, j]] = Quantile[meansMatrix[[i, j]], .1];
  meansQ3[[i, j]] = Quantile[meansMatrix[[i, j]], .9];
  stdevsVals[[i, j]] = Mean[stdevsMatrix[[i, j]]];
  stabilityVals[[i, j]] = Mean[stabilityMatrix[[i, j]]];
  propExt[[i, j]] = Total[extMatrix[[i, j]] / maxsims];
  cdVals[[i, j]] = Mean[cdMatrix[[i, j]]];
  cdQ1[[i, j]] = Quantile[cdMatrix[[i, j]], .1];
  cdQ3[[i, j]] = Quantile[cdMatrix[[i, j]], .9],
  {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*TROPICAL*)

```

```

meansMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stdevsMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stabilityMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
extMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
cdMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];

(*averaging over all timesteps before extinction within a simulation*)
Do[If[extTimeT[[i, j, 1, l]] < tmax, extMatrixT[[i, j, l]] = 1, extMatrixT[[i, j, l]] = 0];
  meansMatrixT[[i, j, l]] = Mean[pT[[i, j, 1, l, 2 ;; extTimeT[[i, j, 1, l]]]]];
  stdevsMatrixT[[i, j, l]] =
    StandardDeviation[pT[[i, j, 1, l, 2 ;; extTimeT[[i, j, 1, l]]]]];
  stabilityMatrixT[[i, j, l]] = meansMatrixT[[i, j, l]] / stdevsMatrixT[[i, j, l]];
  {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

Do[cdMatrixT[[i, j, l]] = CDindex[p[[i, j, 1, l, 2 ;; extTimeT[[i, j, 1, l]]] + .1],
  {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]
Do[cdMatrixT[[i, j]] = DeleteAnomalies[cdMatrixT[[i, j]],
  {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*averaging across simulations with the same params*)
meansValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ1T = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ3T = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stdevsValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stabilityValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
propExtT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ1T = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ3T = ConstantArray[0, {Length[γvals], Length[μTvals]}];

Do[meansValsT[[i, j]] = Mean[meansMatrixT[[i, j]]];
  meansQ1T[[i, j]] = Quantile[meansMatrixT[[i, j]], .1];
  meansQ3T[[i, j]] = Quantile[meansMatrixT[[i, j]], .9];
  stdevsValsT[[i, j]] = Mean[stdevsMatrixT[[i, j]]];
  stabilityValsT[[i, j]] = Mean[stabilityMatrixT[[i, j]]];
  propExtT[[i, j]] = Total[extMatrixT[[i, j]] / maxsims];
  cdValsT[[i, j]] = Mean[cdMatrixT[[i, j]]];
  cdQ1T[[i, j]] = Quantile[cdMatrixT[[i, j]], .1];
  cdQ3T[[i, j]] = Quantile[cdMatrixT[[i, j]], .9],
  {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*create plotting arrays*)
means23 = Transpose[{γvals, meansValsT[[All, 1]]}];
means24 = Transpose[{γvals, meansValsT[[All, 2]]}];

```

```
means25 = Transpose[{γvals, meansVals[[All, 3]]}];
means26 = Transpose[{γvals, meansVals[[All, 4]]}];
```

```
means23Q1 = Transpose[{γvals, meansQ1[[All, 1]]}];
means23Q3 = Transpose[{γvals, meansQ3[[All, 1]]}];
means24Q1 = Transpose[{γvals, meansQ1[[All, 2]]}];
means24Q3 = Transpose[{γvals, meansQ3[[All, 2]]}];
means25Q1 = Transpose[{γvals, meansQ1[[All, 3]]}];
means25Q3 = Transpose[{γvals, meansQ3[[All, 3]]}];
means26Q1 = Transpose[{γvals, meansQ1[[All, 4]]}];
means26Q3 = Transpose[{γvals, meansQ3[[All, 4]]}];
```

```
cd23 = Transpose[{γvals, cdVals[[All, 1]]}];
cd24 = Transpose[{γvals, cdVals[[All, 2]]}];
cd25 = Transpose[{γvals, cdVals[[All, 3]]}];
cd26 = Transpose[{γvals, cdVals[[All, 4]]}];
```

```
cd23Q1 = Transpose[{γvals, cdQ1[[All, 1]]}];
cd23Q3 = Transpose[{γvals, cdQ3[[All, 1]]}];
cd24Q1 = Transpose[{γvals, cdQ1[[All, 2]]}];
cd24Q3 = Transpose[{γvals, cdQ3[[All, 2]]}];
cd25Q1 = Transpose[{γvals, cdQ1[[All, 3]]}];
cd25Q3 = Transpose[{γvals, cdQ3[[All, 3]]}];
cd26Q1 = Transpose[{γvals, cdQ1[[All, 4]]}];
cd26Q3 = Transpose[{γvals, cdQ3[[All, 4]]}];
```

```
propExt23 = Transpose[{γvals, propExt[[All, 1]]}];
propExt24 = Transpose[{γvals, propExt[[All, 2]]}];
propExt25 = Transpose[{γvals, propExt[[All, 3]]}];
propExt26 = Transpose[{γvals, propExt[[All, 4]]}];
```

```
means23T = Transpose[{γvals, meansValsT[[All, 1]]}];
means24T = Transpose[{γvals, meansValsT[[All, 2]]}];
means25T = Transpose[{γvals, meansValsT[[All, 3]]}];
means26T = Transpose[{γvals, meansValsT[[All, 4]]}];
```

```
means23Q1T = Transpose[{γvals, meansQ1T[[All, 1]]}];
means23Q3T = Transpose[{γvals, meansQ3T[[All, 1]]}];
means24Q1T = Transpose[{γvals, meansQ1T[[All, 2]]}];
means24Q3T = Transpose[{γvals, meansQ3T[[All, 2]]}];
means25Q1T = Transpose[{γvals, meansQ1T[[All, 3]]}];
means25Q3T = Transpose[{γvals, meansQ3T[[All, 3]]}];
means26Q1T = Transpose[{γvals, meansQ1T[[All, 4]]}];
means26Q3T = Transpose[{γvals, meansQ3T[[All, 4]]}];
```

```

cd23T = Transpose[{γvals, cdValsT[All, 1]}];
cd24T = Transpose[{γvals, cdValsT[All, 2]}];
cd25T = Transpose[{γvals, cdValsT[All, 3]}];
cd26T = Transpose[{γvals, cdValsT[All, 4]}];

cd23Q1T = Transpose[{γvals, cdQ1T[All, 1]}];
cd23Q3T = Transpose[{γvals, cdQ3T[All, 1]}];
cd24Q1T = Transpose[{γvals, cdQ1T[All, 2]}];
cd24Q3T = Transpose[{γvals, cdQ3T[All, 2]}];
cd25Q1T = Transpose[{γvals, cdQ1T[All, 3]}];
cd25Q3T = Transpose[{γvals, cdQ3T[All, 3]}];
cd26Q1T = Transpose[{γvals, cdQ1T[All, 4]}];
cd26Q3T = Transpose[{γvals, cdQ3T[All, 4]}];

propExtT23 = Transpose[{γvals, propExtT[All, 1]}];
propExtT24 = Transpose[{γvals, propExtT[All, 2]}];
propExtT25 = Transpose[{γvals, propExtT[All, 3]}];
propExtT26 = Transpose[{γvals, propExtT[All, 4]}];

```

Create Individual Plots

Code for each individual subplot. Outputs repressed for processing time.

In[150]:=

```

meansPlot = ListLinePlot[
  {means23, means23Q1, means23Q3, means24, means24Q1, means24Q3, means25,
    means25Q1, means25Q3, means26, means26Q1, means26Q3, l23t, l24t, l25t, l26t},
  PlotStyle → {Lighter[Red], None, None, Red, None, None, Darker[Red],
    None, None, Black, None, None, {Dashed, Lighter[Red]},
    {Dashed, Red}, {Dashed, Darker[Red]}, {Dashed, Black}},
  PlotRange → {{-0.02, 2.02}, {400, 900}}, FillingStyle → Opacity[0.1],
  Filling →
    {1 → {2}, 1 → {3}, 4 → {5}, 4 → {6}, 7 → {8}, 7 → {9}, 10 → {11}, 10 → {12}},
  Frame → {True, True, False, False}, FrameStyle → 14,
  FrameLabel → {None(*"Spectral Exponent γ"*), "Mean Population Abundance"},
  LabelStyle → Directive[Black], ImageSize → 500] ;(*subplot a: temp means*)

stabilityPlot =
  ListLinePlot[{cd26, cd26Q1, cd26Q3, cd25, cd25Q1, cd25Q3, cd24, cd24Q1, cd24Q3,
    cd23, cd23Q1, cd23Q3}, PlotStyle → {Black, None, None, Darker[Red], None, None,
    Red, None, None, Lighter[Red], None, None}, FillingStyle → Opacity[0.1],
  Filling → {10 → {11}, 10 → {12}, 7 → {8}, 7 → {9}, 4 → {5}, 4 → {6}, 1 → {2}, 1 → {3}},
  FillingStyle → Opacity[0.1], PlotRange → {{-0.02, 2.02}, {-0.01, .55}},

```

```

Frame → {True, True, False, False}, FrameStyle → 14,
FrameLabel → {(*"Spectral Exponent  $\gamma$ ",*)None, "Consecutive Disparity Index"},
LabelStyle → Directive[Black], ImageSize → 510];(*subplot b: stability*)

extinctionPlot = ListLinePlot[{propExt23, propExt24, propExt25, propExt26},
  PlotStyle → {Lighter[Red], Red, Darker[Red], Black},
  PlotRange → {{-0.02, 2.02}, Automatic},
  Frame → {True, True, False, False}, FrameStyle → 14,
  FrameLabel → {"Spectral Exponent  $\gamma$ ", "Probability of Extinction"},
  LabelStyle → Directive[Black], ImageSize → 500];
(*subplot c: temp extinction*)

meansPlotT =
  ListLinePlot[{means23T, means23Q1T, means23Q3T, means24T, means24Q1T, means24Q3T,
    means25T, means25Q1T, means25Q3T, means26T, means26Q1T, means26Q3T, lt23t,
    lt24t, lt25t, lt26t}, PlotStyle → {Lighter[Red], None, None, Red, None,
    None, Darker[Red], None, None, Black, None, None, {Dashed, Lighter[Red]},
    {Dashed, Red}, {Dashed, Darker[Red]}, {Dashed, Black}},
  PlotRange → {{-0.02, 2.02}, {360, 850}}, FillingStyle → Opacity[0.1],
  Filling → {1 → {2}, 1 → {3}, 4 → {5}, 4 → {6}, 7 → {8}, 7 → {9}, 10 → {11}, 10 → {12}},
  Frame → {True, True, False, False}, FrameStyle → 14,
  (*FrameLabel → {(*None,*)"Spectral Exponent  $\gamma$ ", "Mean Population Abundance"},*)
  LabelStyle → Directive[Black], ImageSize → 500];(*subplot d: trop means*)

stabilityPlotT =
  ListLinePlot[{cd23T, cd23Q1T, cd23Q3T, cd24T, cd24Q1T, cd24Q3T, cd25T, cd25Q1T,
    cd25Q3T, cd26T, cd26Q1T, cd26Q3T}, PlotStyle → {Lighter[Red], None,
    None, Red, None, None, Darker[Red], None, None, Black, None, None},
  PlotRange → {{-0.02, 2.02}, {-.01, .55}}, FillingStyle → Opacity[0.1],
  (*PlotLabelStyle["Specialist",16],*)
  Filling → {10 → {11}, 10 → {12}, 7 → {8}, 7 → {9}, 4 → {5}, 4 → {6}, 1 → {2}, 1 → {3}},
  Frame → {True, True, False, False}, FrameStyle → 14,
  (*FrameLabel → {None(*"Spectral Exponent  $\gamma$ "*), "Population Stability"},*)
  LabelStyle → Directive[Black], ImageSize → 510];(*subplot e: trop stability*)

extinctionPlotT = ListLinePlot[{propExtT23, propExtT24, propExtT25, propExtT26},
  PlotStyle → {Lighter[Red], Red, Darker[Red], Black},
  PlotRange → {{-0.02, 2.02}, Automatic},
  Frame → {True, True, False, False}, FrameStyle → 14,
  FrameLabel → {"Spectral Exponent  $\gamma$ ", None(*"Probability of Extinction"*)},
  LabelStyle → Directive[Black], ImageSize → 500];(*subplot f: trop extinction*)

```

Fig 4: Extinction Metrics

Create Fig 4. Labeling added in Adobe Illustrator 2024.

In[157]:=

```
fig4 = GraphicsColumn[{meansPlot, stabilityPlot, extinctionPlot, meansPlotT,
  stabilityPlotT, extinctionPlotT}, Spacings → {0, 0}, ImageSize → 500]
(*fig4=GraphicsGrid[{{meansPlot,meansPlotT},{stabilityPlot,stabilityPlotT},
  {extinctionPlot,extinctionPlotT}},Spacings→{0,0},ImageSize→1000]*)

Export[saveto<>"Figure4.pdf", fig4, ImageResolution → 1000];
(*Export[saveto<>"meansPlot.pdf",meansPlot,ImageResolution→1000]
  Export[saveto<>"meansPlotT.pdf",meansPlotT,ImageResolution→1000]
  Export[saveto<>"stabilityPlot.pdf",stabilityPlot,ImageResolution→1000]
  Export[saveto<>"stabilityPlotT.pdf",stabilityPlotT,ImageResolution→1000]
  Export[saveto<>"extinctionPlot.pdf",extinctionPlot,ImageResolution→1000]
  Export[saveto<>"extinctionPlotT.pdf",extinctionPlotT,ImageResolution→1000]*)
```

Out[157]=





